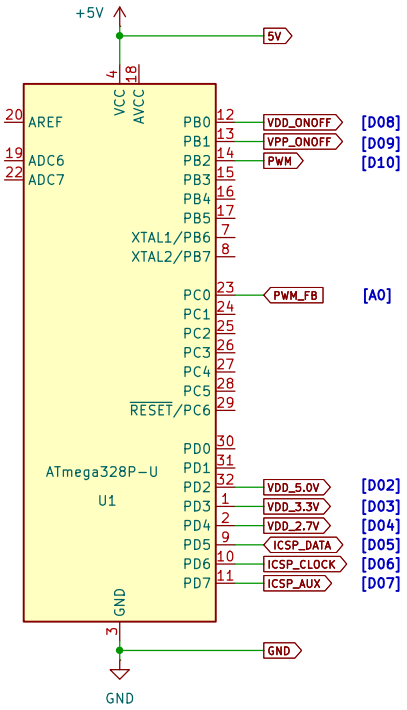
PicArd Programmer Hardware – Revision 3

Arduino UNO R3 (ATMEGA328PU 32-lead TQFP)



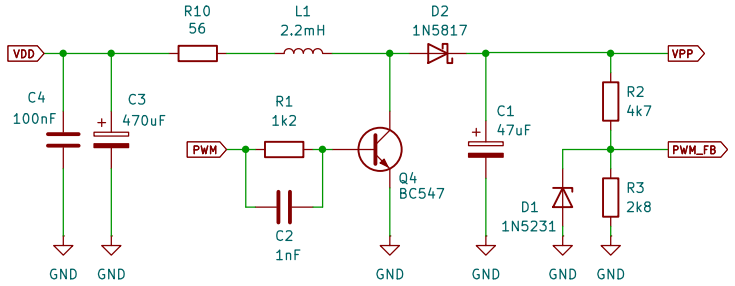
Notes:

a) Showing only relevant connections
b) Into the brackets: Arduino Uno R3 board silk screen markings.

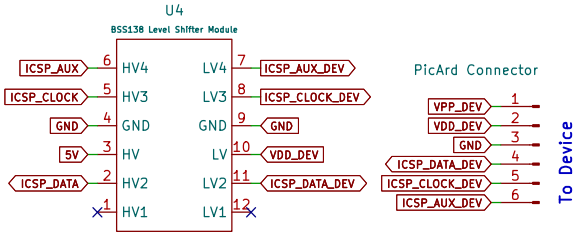
Table 1 – Labels Descriptions

Label	Description
5V	Arduino UNO R3 5V (from USB)
3V3	Positive 3.3V (from Arduino Uno)
PWM	Pulse with modulation to DC/DC Booster
PWM_FB	PWM voltage feedback
VDD_ONOFF	Turns on/off VDD to device
VPP_ONOFF	Turns on/off VPP to device
VDD_5.0V	Select/Unselect 5.0V as VDD
VDD_3.3V	Select/Unselect 3.3V as VDD
VDD_2.7V	Select/Unselect 2.7V as VDD
ICSP_DATA	In-circuit serial programming (ICSP) data pin [0V to 5V]
ICSP_DATA_DEV	ICSP data pin to device [0V to VDD_DEV]
ICSP_CLOCK	ICSP clock pin [0V to 5V]
ICSP_CLOCK_DEV	ICSP clock pin to device [0V to VDD_DEV]
ICSP_AUX	ICSP for Low Voltage Programming (LVP) [0V-5V]
ICSP_AUX_DEV	ICSP for LVP [0V to VDD_DEV]
VDD	Selected VDD
VDD_DEV	Device VDD Pin
VPP	Programming voltage
VPP_DEV	Device VPP Pin
GND	Ground

DC/DC Boost

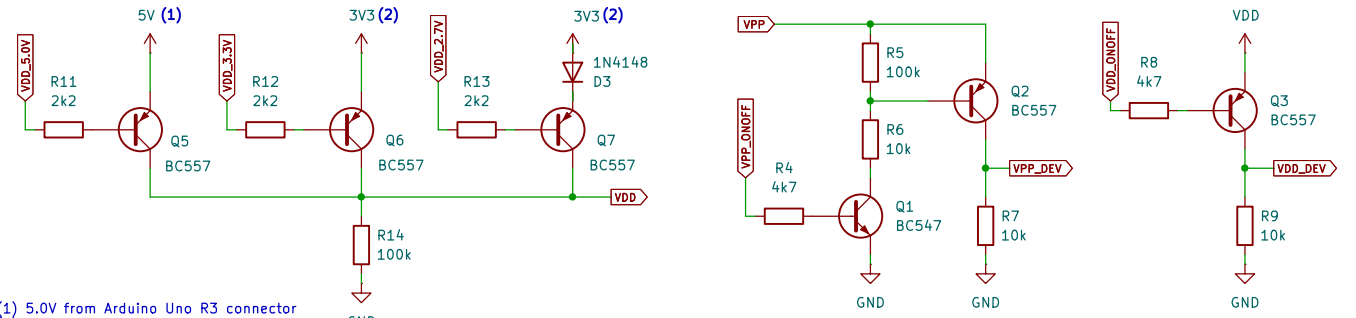


Level-Shifters and Connector



If there are communication issues with this level-shifter module, use weaker pull-ups (see below)

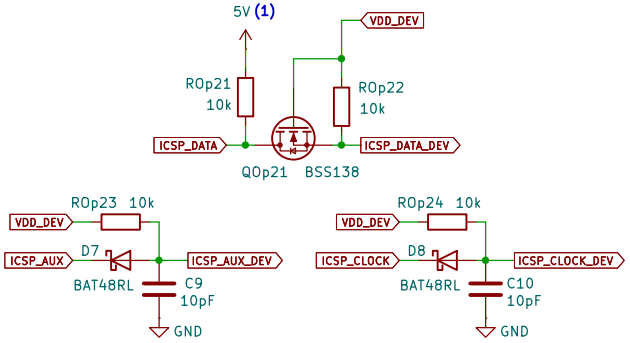
VPP and VDD control



(1) 5.0V from Arduino Uno R3 connector
(2) 3.3V from Arduino Uno R3 connector

Option: Replace Level Shift Module with BSS138 for ICSP_DATA and diodes for ICSP_AUX and ICSP_CLOCK

Use ROp2x values between 2k and 4k7 if there are communication issues.



Level-Shifter weaker pull-ups

Use RWx values between 2k and 4k7 if there are communication issues using Level Shifter module



Revision 2 and 3 can be usaed for new projects
Only revision 3 allows reading/flashing PICs that operate below 5V
Revision 3
https://github.com/NelsonBittencourt/PicArd/
Nelson Rossi Bittencourt

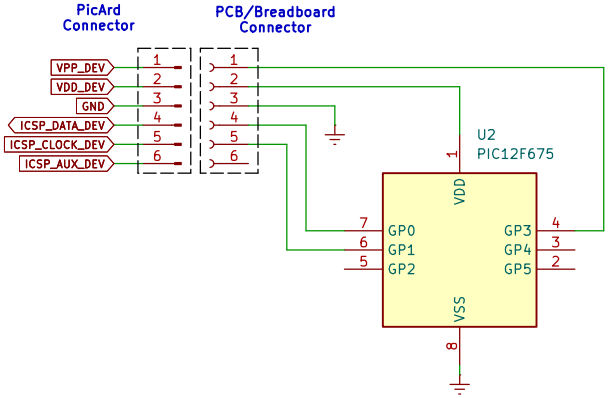
Sheet: /picard_hardware/
File: picard_hardware.kicad_sch

Title: PicArd Programmer

Size: A4	Date: 2026-11-11	**Rev: 3**
KiCad E.D.A. 10.0.6		Id: 1/3

PicArd Programmer – Connection Examples

Example 1: PIC12F675 (PDIP, SOIC, DFN-S)

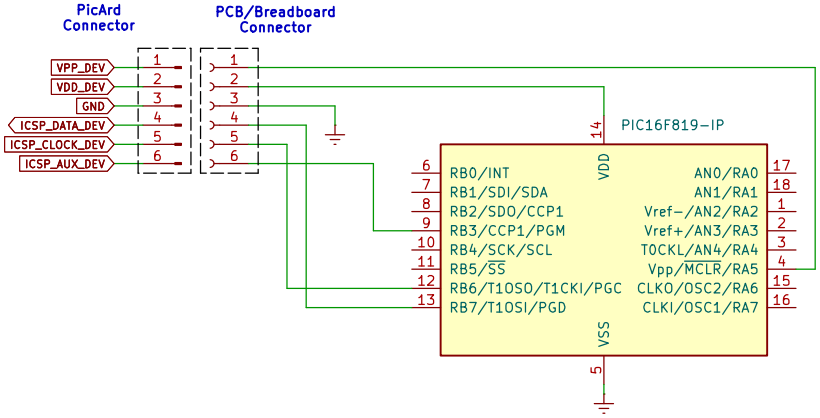


Notes:

a) Connections according to "PIC12F629/675 Memory Programming Specification" (https://ww1.microchip.com/downloads/en/DeviceDoc/41173c.pdf);

b) PIC12F675 can only be programmed via HVP, ICSP_AUX not used.

Example 2: PIC16F819 (DIP, SOIC)



Notes:

a) Connections according to "PIC16F818/819 Memory Programming Specification" (https://ww1.microchip.com/downloads/en/DeviceDoc/39603c.pdf);

b) PIC16F818/819 supports LVP and HVP modes. Selection is made via software.

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Revision 3
https://github.com/NelsonBittencourt/PicArd/
Nelson Rossi Bittencourt

Sheet: /picard_hardware/Picard_Connection_Example/
File: picard_hardware_connection_examples.kicad_sch

Title: PicArd Programmer – Connection Examples

Size: A4	Date: 2026-11-11	**Rev: 3**
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PicArd Programmer – Typical voltages and signals

Fig. A: PWM Signal

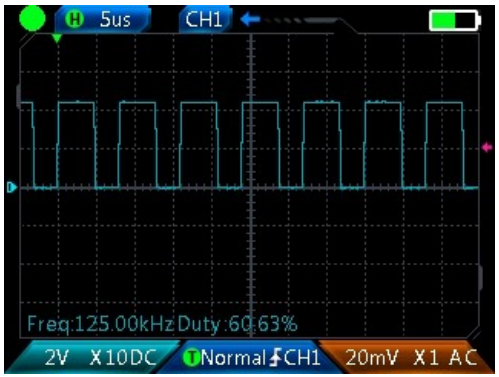


Fig. B: PWM in Q10 collector

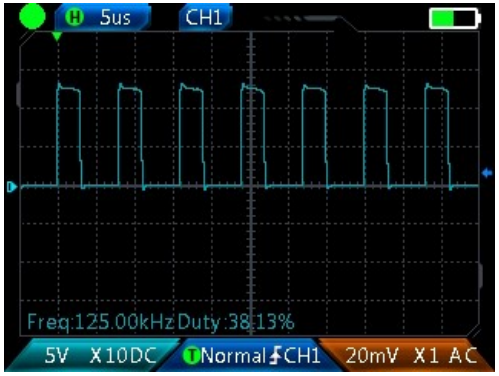
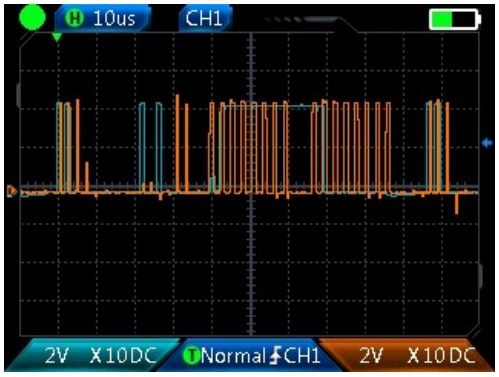
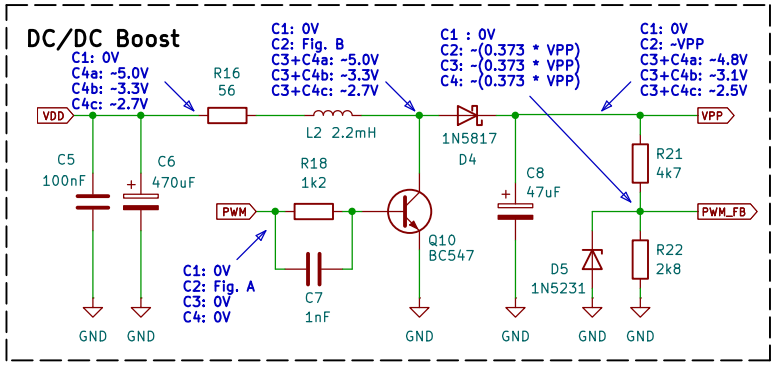
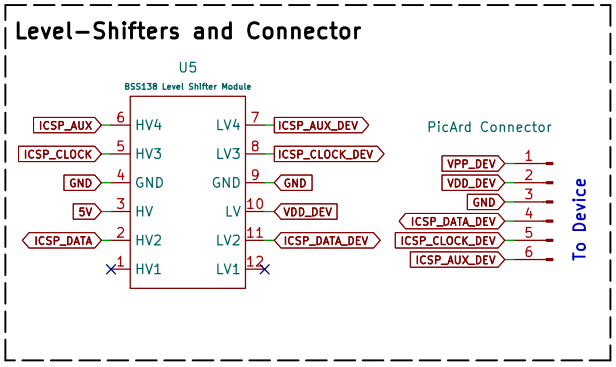


Fig. C: ICSP_DATA (blue) & ICSP_CLOCK







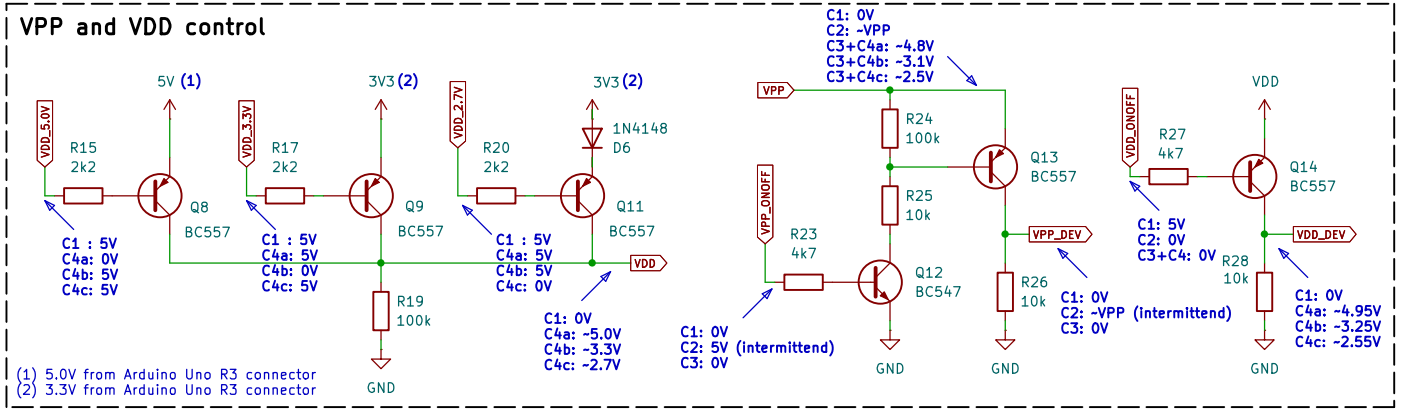


Table 2 – Condition Description	
Condition #	**Description**
C1	Arduino Uno connected in USB port/GUI connected or not
C2	HVP mode
C3	LVP mode
C4a	5.0V VDD Device
C4b	3.3V VDD Device
C4c	2.7V VDD Device

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Sheet: /Picard_Voltages/
File: picard_voltages.kicad_sch

Title: PicArd Programmer – Typical voltages & signals

Size: A4	Date: 2026-11-11	**Rev: 3**
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